# Exercise 5: Turning Flight

A turbojet aircraft (symmetric quadratic drag polar) begins to glide at a height $h_0$=1000 $m$ above the ground.

| | | |
|---|---|---|
| Zero-lift drag coefficient | $C_{D0}$ | 0.031 |
| K-factor | $k$ | 0.026 |
| Reference area | $S_{ref}$ | 18 $m^2$ |
| Aircraft mass | $m$ | 2000 kg |
| Maximum engine performance parameter | $L_P$ | 6000 |

| | | |
|---|---|---|
| Height | h | 1000 m |
| Density | $\rho$ | $1.1117 \frac{\mathrm{kg}}{\mathrm{m}^3}$ |

a) What is the maximum horizontal distance the aircraft can cover?

b) What are the control inputs to the point mass model and how can the pilot influence them?

c) Derive the required velocity for achieving the maximum climb angle. Which climb angle and which velocity is reached?

d) The aircraft now flies a turn (see the data below). Which kinematic bank angle and what velocity is required for this maneuver?

| | | |
|---|---|---|
| Height and density | h / $\rho$ | $1000\ \mathrm{m} / 1.1117 \frac{\mathrm{kg}}{\mathrm{m}^3}$ |
| Turn rate | $\dot{\chi}$ | $5 \frac{^\circ}{\mathrm{s}}$ |
| Lift coefficient during turn | $C_{L,turn}$ | 1 |